



Automated Fish Health Monitoring Platform

- Distributed imaging nodes for aquatic research tanks
- Water quality monitoring and rack level control
- Cloud based AI phenotype analysis
- Wesley Smith | Technical Presentation

The Problem



Thousands of tanks require continuous monitoring

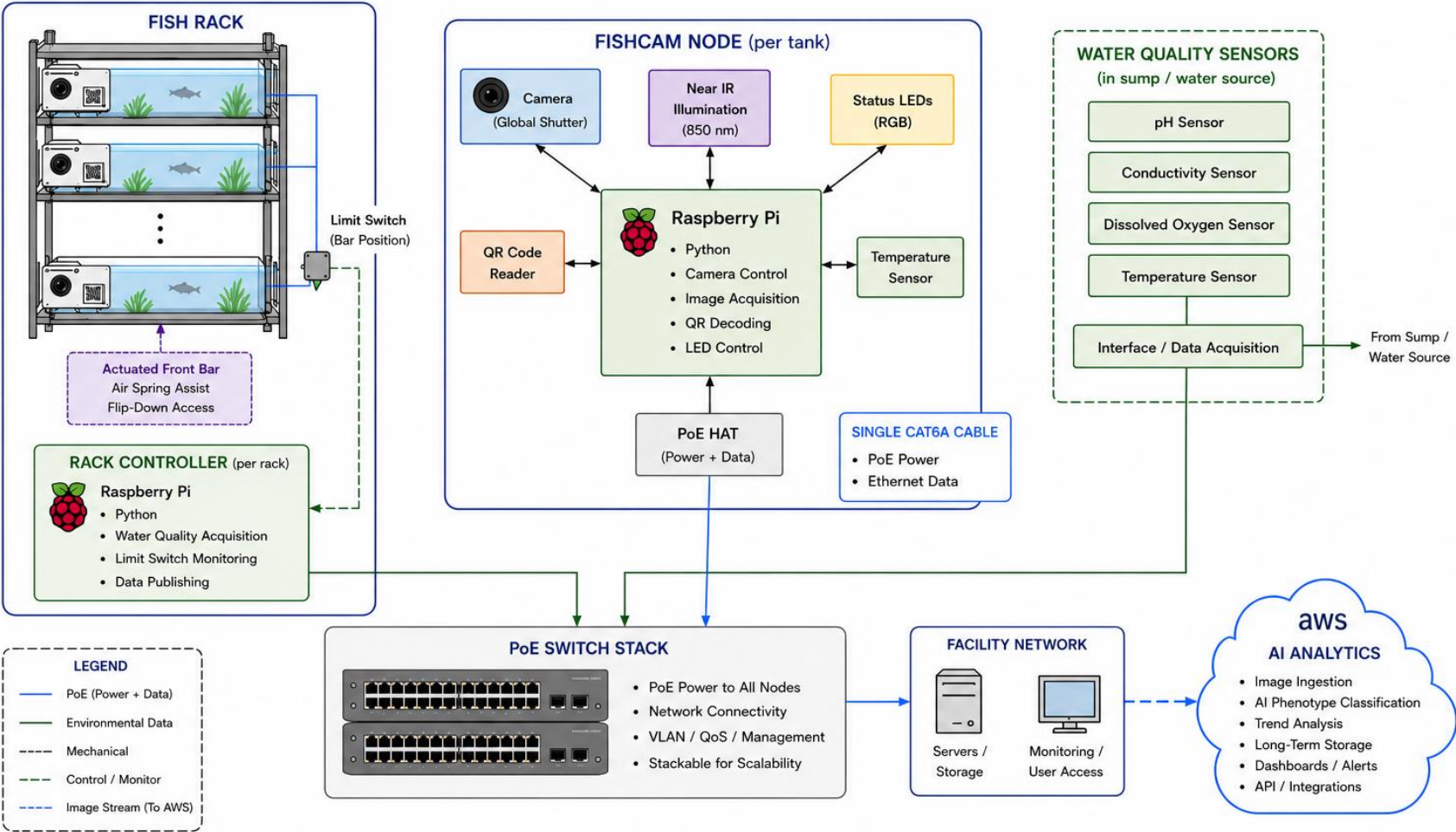
- Count fish without relying on manual observation
- Identify phenotypes: healthy, diseased, dying, or dead
- Monitor water quality alongside health trends
- Reduce labor and improve consistency across racks
- Create a scalable data stream for AI analysis

Overall System Architecture

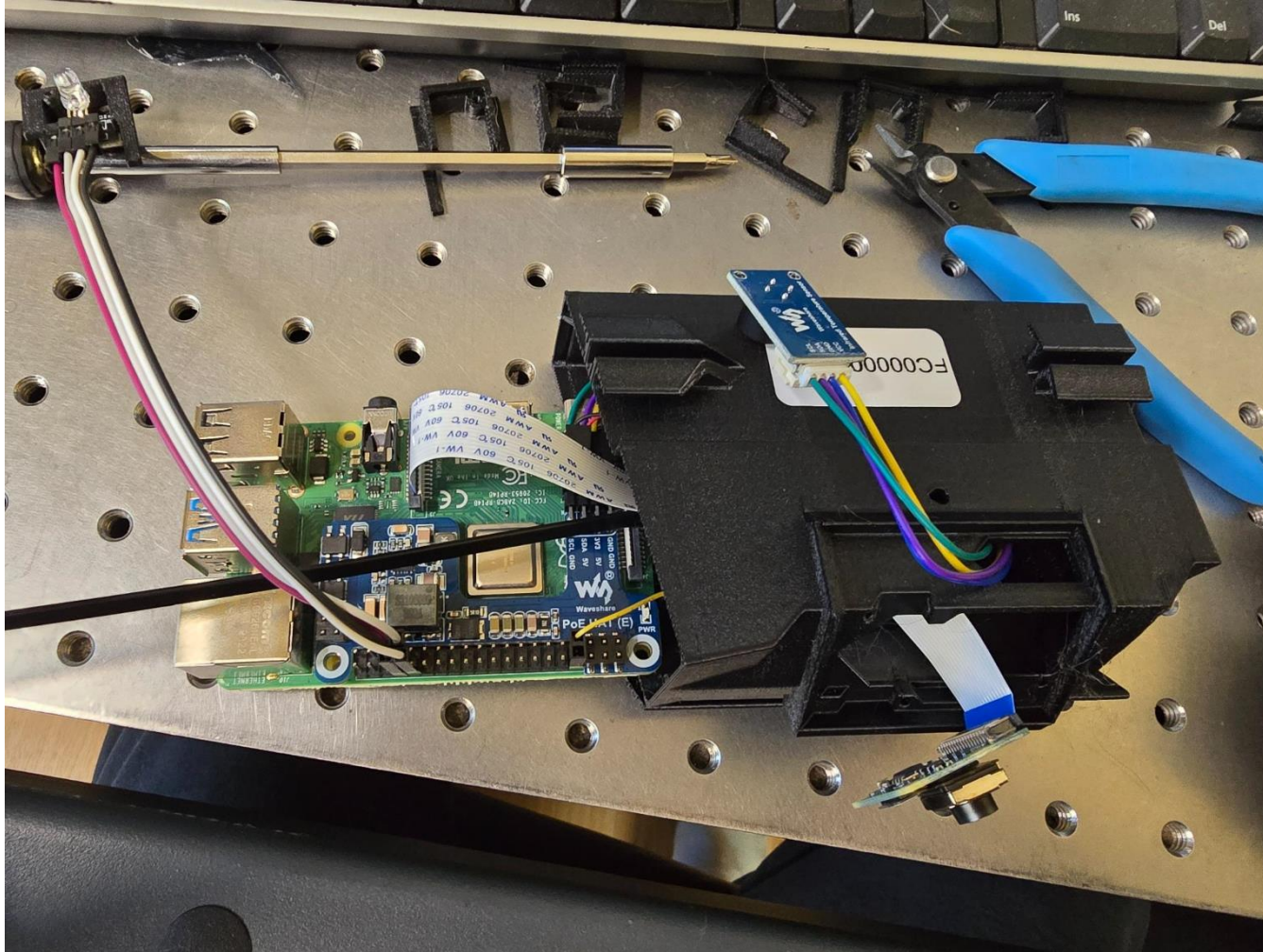
Distributed FishCam nodes, rack level water quality monitoring, PoE networking, and AWS AI analytics

OVERALL SYSTEM ARCHITECTURE

*Distributed FishCam Nodes with Water Quality Monitoring
Powered and Connected over PoE Network*



Engineering Challenges



What had to work together

- Optics, electronics, mechanics, and software in one compact package
- Reliable imaging in a wet laboratory environment
- PoE power, networking, and embedded processing
- Repeatable physical interfaces for deployment and service
- Automated validation before units reached the rack

Challenge 1: Camera Placement



Front mounting was the only viable optical path

- Top: air water interface caused distortion and condensation issues
- Side: insufficient clearance between adjacent tanks
- Bottom: debris accumulated in the field of view
- Rear: tank baffles interrupted line of sight
- Front: repeatable, unobstructed image geometry

- Selected front placement, then designed the mechanical access system around that constraint.

Challenge 2: Customer Access

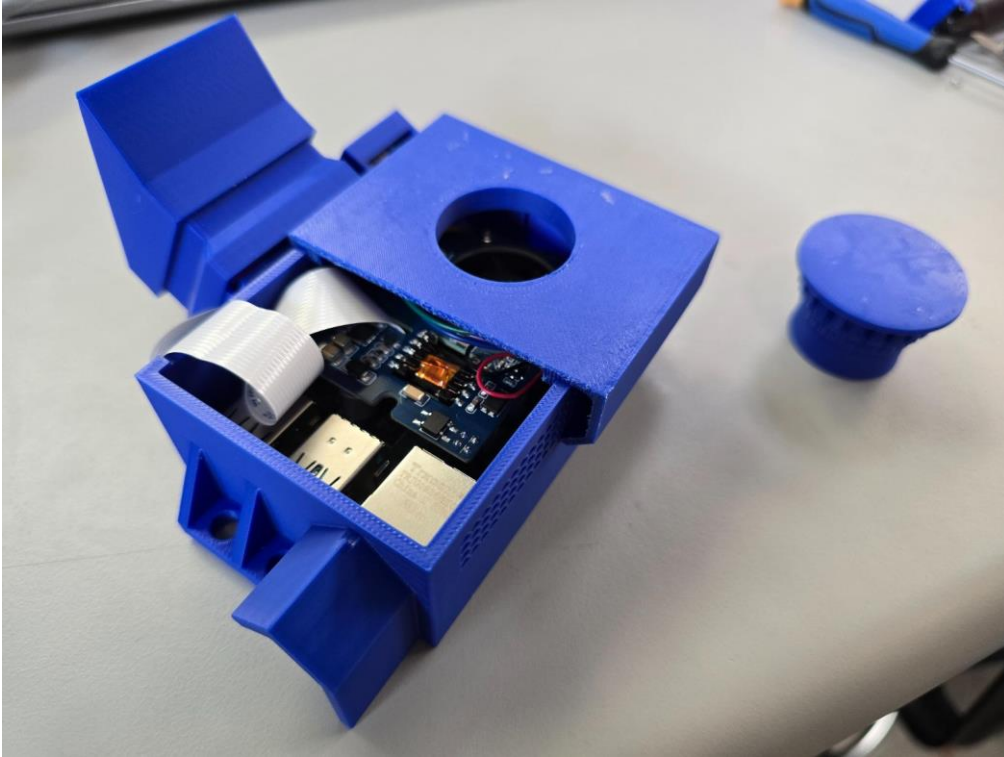


Access without losing alignment

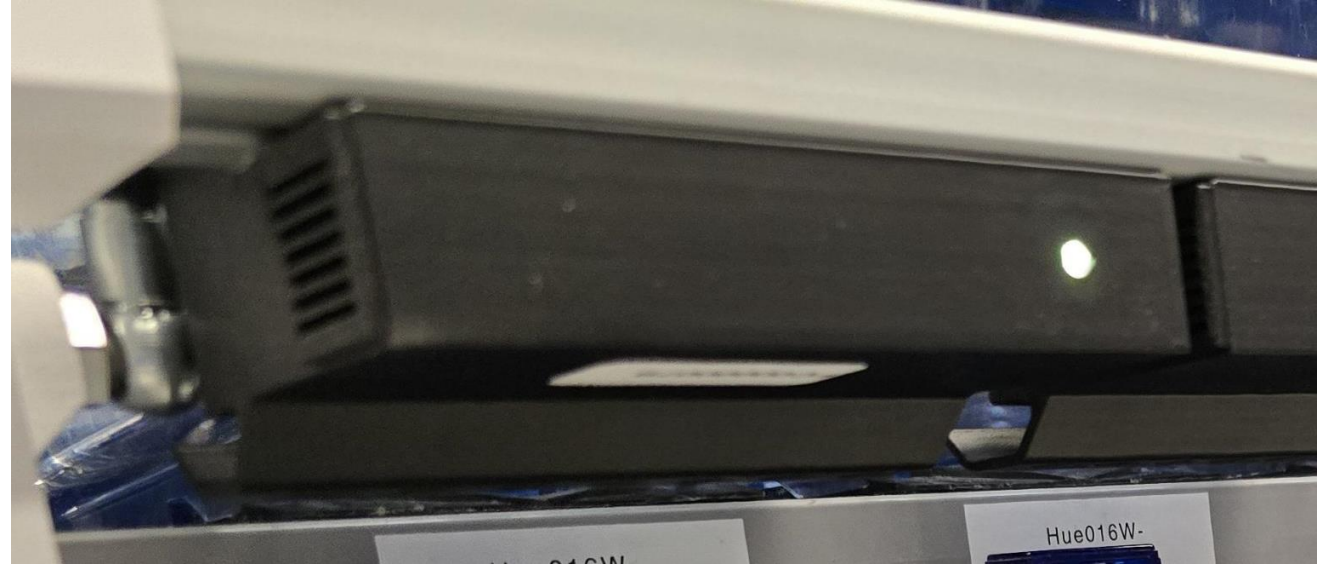
- Flip down front bar provided tank access without removing cameras
- Gas spring reduced opening force and held the bar open
- Mechanical return preserved camera position after every use
- Limit switch reported bar position to the rack controller
- Image acquisition paused while the bar was open

Challenge 3: Thermal Management

Early design: top vent



Final design: side louvers + miniature fan



- Early top vent solved cooling but created a water ingress risk.
- Final design used a miniature fan and side grating to maintain airflow while rejecting direct splashes.
- Result: improved environmental robustness without giving up thermal performance.

Challenge 4: Environmental Monitoring



Probe manifold



Sensor integration



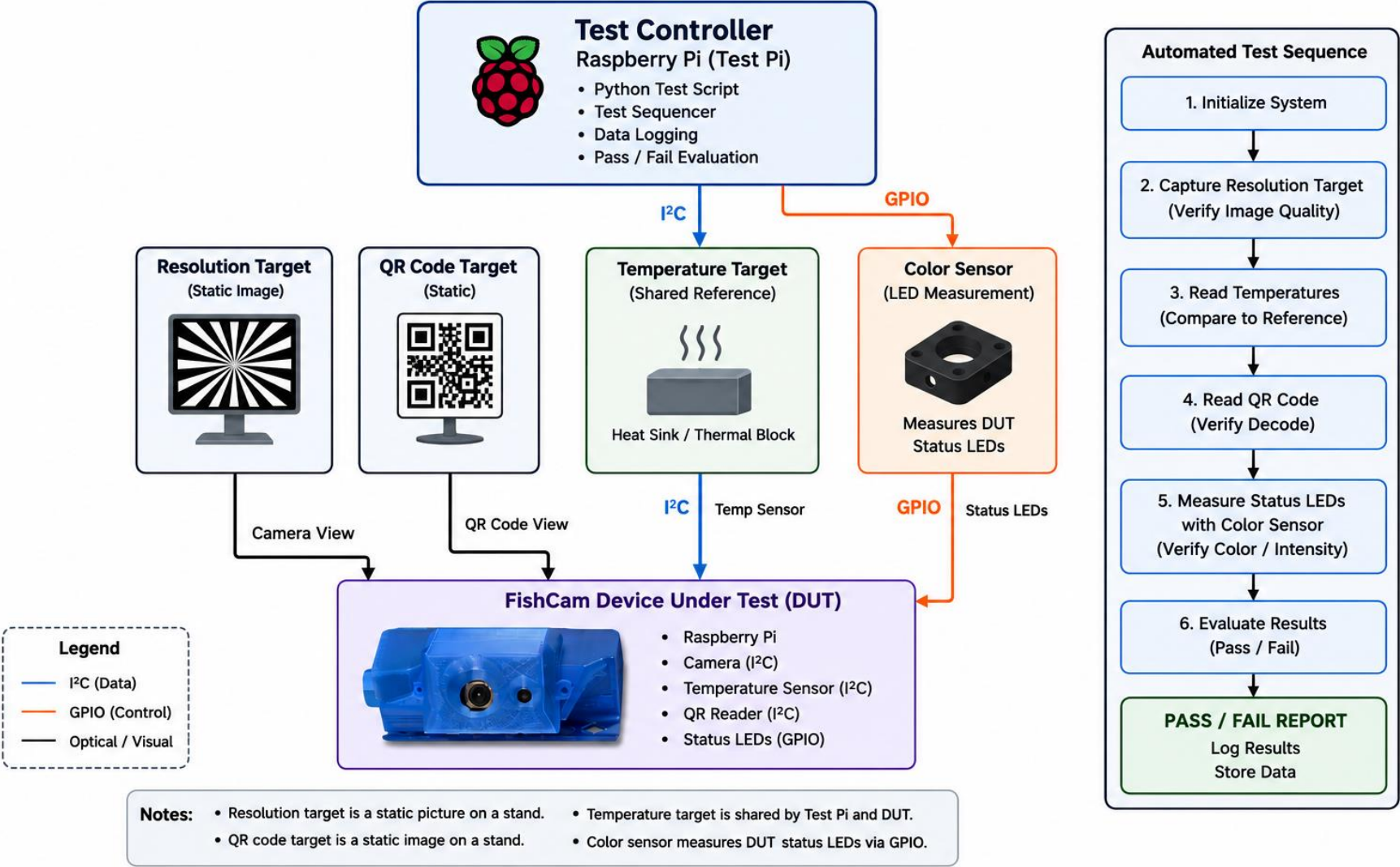
Installed in sump

- Custom probe manifold integrated pH, conductivity, dissolved oxygen, and temperature sensors.
- Material selection mattered: the printed plastic had to avoid leaching into circulating water.
- Consistent sensor positioning supported repeatable environmental measurements.

Automated Production Test

Off rack bench validation before deployment

AUTOMATED BENCH TEST SYSTEM





Platform Outcomes

- Deployed multi tank monitoring platform
- One FishCam node installed per tank
- Automated AI image analysis
- Continuous environmental monitoring
- Automated production validation

Systems Engineering

Requirements evolved through integration and field testing.

Design for Serviceability

Maintenance access influenced multiple architectural decisions.

Prototype Early

Rapid iteration uncovered optical, thermal, and environmental issues before production.

Automate Validation

Automated testing improved manufacturing consistency and reduced test time.

Questions?

